Cody Kochmann

Principal Infrastructure Engineer, Security Developer & DevSecOps Enthusiast

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Summary

I'm a principal infrastructure engineer, security developer, and DevSecOps enthusiast. Whether you need to accelerate your developer workflows through automation, harden your infrastructure against state-sponsored attacks, or design high-performance distributed systems, I have a proven track record of driving mission-critical company goals.

Technical Skills

- **Infrastructure & Cloud:** Kubernetes (EKS, AKS, GKE), Terraform, AWS, Azure, Linux, CloudFormation, ArgoCD, Docker
- **Security & Operations:** DevSecOps, Istio, Falco, Vault, Trivy, Threat Detection, Vulnerability Management, RBAC, Grafana, ELK/HELK Stack
- **Languages & Automation:** Python, Golang, Rust, Bash, CI/CD (GitLab, GitHub Actions, Buildkite), Infrastructure as Code (IaC)

Experience

NICO - 2025 to Present

Roles: Lead Infrastructure Engineer

Programming Languages: Bash, Python, YAML

Technologies: Azure, Kubernetes (AKS), Terraform, ArgoCD, Argo Workflows, Grafana Stack, Istio, Falco, HashiCorp Vault (OpenBao), SealedSecrets, Zot, Trivy, Agentic AI (Claude, Hermes)

Achievements:

- **Security & Operations:** Hardened network traffic by deploying an Istio service mesh, configured Falco for real-time intrusion detection (IDS), and established strict RBAC across all environments.
- **Supply Chain Security:** Secured the software supply chain by building isolated build environments, integrating Trivy for automated container vulnerability scanning, and managing the Zot registry. Safely managed credentials utilizing Vault and SealedSecrets.
- **Cloud Infrastructure:** Lead the comprehensive overhaul of the Third-Party Risk Management (TPRM) Azure infrastructure, establishing a mature, highly resilient posture rooted in industry best practices.
- **Kubernetes & IaC:** Architected and executed the migration of all live infrastructure to Kubernetes, fully codifying the environment into modular, reusable Terraform components.
- **AI-Accelerated Operations:** Leveraged Agentic AI tooling as a force multiplier to rapidly architect, test, and deploy infrastructure as code, significantly accelerating project delivery timelines.
- **GitOps & Automation:** Championed the adoption of immutable infrastructure principles by implementing GitOps workflows with ArgoCD.
- **Observability & Monitoring:** Designed and deployed centralized observability and logging pipelines using the complete Grafana stack.

Independent AI Researcher - 2024

Roles: Independent Researcher, AI Consultant

Programming Languages: Python, Bash

Technologies: Large Language Models (LLMs), Agentic Frameworks, Prompt Engineering, Vector Databases

Achievements:

- Designed and tested secure sandboxing methods to safely provide LLM capabilities within highly sensitive and restricted environments.
- Analyzed the security risks and hallucination vectors of deploying LLMs in enterprise environments, developing strategies for safe adoption.
- Assisted in critical research initiatives focused on optimizing AI operational spend and reducing infrastructure overhead.
- Researched and implemented local GPU optimization techniques to maximize the performance of on-premise inferencing.
- Mastered advanced prompting techniques and agentic workflows (utilizing tools like Antigravity, Claude and Ollama) which directly translated into immense productivity multipliers for future infrastructure automation roles.
- Conducted extensive independent and collaborative research into Large Language Model (LLM) design, training methodologies, and practical utilization for software engineering.

Strake - September 2023 to December 2023

Strake closed business in December 2023.

Roles: Principal Infrastructure Engineer

Programming Languages: Make, Python, Bash

Technologies: GitHub, CloudFormation, CodeDeploy, BuildKite, Postgres, SQLite3, SQS, Lambda

Achievements:

- Drove the development of the tools and workflows needed to generate ephemeral development environments for engineers to safely preform isolated experiments.
- Designed and implemented tools that provided deep insights to every corner of the company's infrastructure, development patterns, and communication effectiveness.

QOMPLX - September 2019 to July 2023

QOMPLX closed business in July 2023.

Roles: Principal Cloud Platform Automation Engineer, Project Lead, Kubernetes SME, Solutions Architect, Security Architect, Security Developer, Mentor

Programming Languages: Python, Golang, Rust, Make, Bash, Jinja, YAML

Technologies: GitLab, CI/CD Pipelines, CI Runners, Kubernetes, Docker, Terraform, Linux Kernel, rsyslog, syslog-ng, Postgres, SQLite3, perf, ebpf, LVM2, Loki, Grafana, Kafka, ElasticSearch, Logstash, Kibana, Kafka, Prometheus, Aqua, Sysdig, Twistlock, CrowdStrike, Linux Kernel, Digital Ocean, Linode, AWS (EKS, ECS, EC2, Lambda, CloudTrail, SQS, S3, Lambda, Config, IAM), Istio, Argo, Debian, Amazon Linux, CoreOS, Ubuntu, Fedora, FreeBSD, CentOS, VMWare

Achievements:

- **Security & Threat Management:** Drove security and vulnerability scanning operations company-wide. Identified and prevented active security breaches and remediated thousands of legacy infrastructure vulnerabilities.
- **Infrastructure & Kubernetes:** Spearheaded the company's comprehensive migration to Kubernetes as a Subject Matter Expert (SME) and Security Architect.
- **Client Operations:** Salvaged major client relationships by directly resolving complex product issues in live production environments and managing the security of specialized on-prem deployments.

- **Automation & CI/CD:** Led the design and development of the company's GitLab organization and CI/CD pipelines, building tools that enabled development teams to securely automate their own deployments.
- **Observability:** Architected centralized log collection systems, providing unprecedented levels of observability across all operational layers of the company.
- **Technical Leadership:** Acted as the platform team's Scrum Master and held weekly lectures for developers on defensive coding, infrastructure optimization, and high-performance architecture.

EMBEDDED FLIGHT SYSTEMS INC (NASA) - January 2017 to July 2019

Roles: Network Operations Engineer, Security Developer, Mentor

Programming Languages: Python, Golang, Rust, Bash

Technologies: Docker, Kubernetes, rocket, Twistlock, Sysdig, Aqua, HELK, Security Onion, OpenShift, Kubernetes, Docker, Amazon Lambda, Amazon ECS, Amazon EKS, Digital Ocean, Google GKE, Systemd, RPM, Debian, Amazon Linux, CoreOS, Ubuntu, Fedora, FreeBSD, CentOS, Cisco, VMWare

Achievements:

- **Security Architecture:** Designed and built a high performance DDOS mitigation tool capable of protecting NASA's network from state sponsored attacks.
- **Security Operations:** Architected a high performance distributed ELK cluster to ingest and analyze network logs across our hybrid cloud / on-prem network. Designed and implemented a multitude of netflow/packet capture/firewall log analysis tools to give the company deeper insight to AWS, on-prem, and container network traffic.
- Preformed security analysis for numerous black box solutions to identify how safe the tools were to use and how we could keep them locked down when risks were identified.
- Lead the efforts for analyzing which container security analysis tools were worthy of NASA's security standards.
- Taught developers about the principals of defensive coding and how to build highly resilient architectures that met NASA's reliability and security standards.

Open Source Development | June 2013 - Present

I am the lead author and architect for all of the following projects.

battle_tested	Fully automated Python fuzzer that highlights every crash a piece of code will raise in production.
generators	High performance pipeline processing library written in pure python.
graphdb	The fastest pure python graph database available on pypi.
queued	Turns functions into fully functional multi-core queued services.
strict_functions	Function decorators for more secure python code execution.
stricttuple	Python tuples with rust style built in rule enforcement
litecollections	python collection types except they're all backed by SQLite

References

All of the following were solid people I had the honor to work with and would love to work with again one day.

Marcin Pohl	Systems / Security Engineer @ NASA	marcin.pohl@nasa.gov
Adam Younce	CTO @ Strake	ayounce@ripcord.net
Jim Treinen	CEO @ Strake	
Beck Norris	Information Security / Compliance Expert @ Frontier Airlines	rrnorris33@gmail.com
Mason Walton	Automation Engineer @ QOMPLX	fin.mwalto7@gmail.com
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